MIL-M-60817A(MU)
19 September 1969

SUPERSEDING
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12 June 1967

MILITARY SPECIFICATION

MODULES, RADAR CHRONOGRAPH: M36 (ENCAPSULATED)

1. SCOPE

1.1 This specification covers eight different types of Modules used in the M36 Radar Chronograph. The Modules are used to provide gating, reset conditions, and internal calibration signals. The Modules covered by this specification are as follows:

Oscillator Doppler Gate (A1/A2 Module)	10525492
Generator, Gate (A3 Module)	10525493
Pulse Generator (A4 Module)	10525494
Amplifier, AF/RF (A5/A9 Module)	10525495
Indicator, Reset (A6 Module)	10525496
Oscillator 95.648 KHz, Radio Freq. (A7 Module)	8215500
Gate Delay (A8 Module)	10525498
Oscillator 75 KHz, Radio Freq. (A10 Module)	8215497

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction, M18 Manufacturing Process and Inspection of

FSC 1290

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

C10525492	Oscillator Doppler Gate (A1/A2 Module)
C10525493	Generator, Gate (A3 Module)
C10525494	Pulse Generator (A4 Module)
C10525495	Amplifier, AF/RF (A5/A9 Module)
C10525496	Indicator, Reset (A6 Module)
C8215500	Oscillator 95.648 KHz, Radio Freq. (A7 Module)
C10525498	Gate Delay (A8 Module)
C8215497	Oscillator 75 KHz, Radio Freq. (A10 Module)

PACKAGING DATA SHEETS

10525492	Oscillator Doppler Gate (A1/A2 Module)
10525493	Generator Gate (A3 Module)
10525494	Pulse Generator (A4 Module)
10525495	Amplifier, AF/RF (A5/A9 Module)
10525496	Indicator, Reset (A6 Module)
8215500	Oscillator 95.648 KHz, Radio Freq. (A7 Module)
10525498	Gate Delay (A8 Module)
8215497	Oscillator 75 KHz, Radio Freq. (A10 Module)

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Fabrication.- The Modules shall be manufactured in accordance with the applicable drawing (see 2.1) and drawings pertaining thereto.

3.2 General specifications.- The Modules shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances

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(c) Inorganic protective surface finishes
(d) Workmanship
(e) Soldering electrical connections

3.2.2 MIL-C-46357

(a) Wiring
(b) Thread fasteners
(c) Marking
(d) Finishes
(e) Etched circuit assemblies

3.3 ENVIRONMENTAL

3.3.1 Shock. - The Modules shall be capable of three (3) shocks of 50 gravity units (g's) of eleven (11) millisecond duration in each of the following directions:

(a) Bottom down
(b) Front down
(c) Right side down
(d) Back down
(e) Left side down
(f) Top down

Subsequent to the above shock, the Modules shall not have been damaged or adversely affected, and shall meet the respective Module requirements of 3.5.

3.3.2 Vibration. - The Modules shall be capable of continuous operation while being vibrated in a simple harmonic motion having an amplitude of 0.03 (0.06-inch maximum total excursion) with the frequency being varied uniformly between the limits of ten (10) and fifty-five (55) hertz. The entire frequency range, from ten(10) to fifty-five(55) hertz and return to ten (10) hertz, shall be uniformly traversed in one (1) minute plus or minus 15 seconds. This motion shall be applied for a period of thirty (30) minutes plus or minus one (1) minute in each of three (3) mutally perpendicular directions (x, y, z axis) for a total of 1.5 hours. During the above vibration, the Modules shall meet the respective Module requirements of 3.5 twice in each plane. At all other times, the Modules shall not have been damaged or adversely affected and shall meet the respective Module requirements of 3.5.

3.3.3 Humidity. - The Modules shall be capable of continuous operation during four (4) hours plus or minus ten (10) minutes of continuous exposure to a relative humidity of ninety (90) to ninety-five (95) per cent at a temperature of plus 100°F plus or minus 5°F. During and subsequent to the above humidity, the Modules shall not have been damaged or adversely affected and shall meet the respective Module requirements of 3.5.

3.3.4 Sand and Dust. - The Modules shall be capable of withstanding three (3) hours of continuous exposure to blowing sand of particule size 0.01 to 1.00mm and blowing dust of particule size 0.0001 to 0.01mm with density

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between 0.1 to 0.5 gram per cubic foot of dust at a temperature of plus 75°F plus or minus 5°F, a relative humidity between fifteen (15) to thirty (30) per cent, and a wind velocity of fifteen (15) knots. Subsequent to the above sand and dust exposure, the Modules shall not have been damaged or adversely affected and shall meet the respective Module requirements of 3.5.

3.3.5 Altitude.

3.3.5.1 Operating. - The Modules shall be capable of continuous operation during four (4) hours of continuous exposure to an altitude of 10,000 feet at a temperature of plus 75°F plus or minus 5°F. During and subsequent to the above exposure, the Modules shall not have been damaged or adversely affected and shall meet the respective Module requirements of 3.5.

3.3.5.2 Non-operating. - The Modules shall be capable of withstanding four (4) hours of continuous exposure to an altitude of 50,000 feet at a temperature of minus 65°F plus or minus 5°F. Subsequent to the above exposure, the Modules shall not have been damaged or adversely affected and shall meet the respective Module requirements of 3.5.

3.3.6 Fungus. - The Modules shall be capable of withstanding twenty-eight (28) days exposure to a composite spore suspension consisting of the following cultures:

(a) Chaetomuim globsom
(b) Aspergillus flavus
(c) Aspergillis versicalor
(d) Penicillum luteum
(e) Rhizopus nigricans

The composite spore suspension shall be incubated at an ambient temperature of 80°F plus or minus 4°F with an RH of 87 plus or minus 2%. Subseuqent to the above exposure, the Modules shall not have been damaged or adversely affected and shall meet the respective Module requirements of 3.5.

3.3.7 Storage Temperatures. - The Modules shall be capable of withstanding temperature stabilization at plus 160°F and minus 80°F for a period of eight (8) hours at each temperature extreme. Subsequent to each of the above temperature extremes (+160°F and -80°F) after temperature stabilization at plus 70°F plus or minus 15°F, the Modules shall not have been damaged or adversely affected and shall meet the respective Module requirements of 3.5.

3.3.8. Operating Temperatures. - The Modules shall be capable of continuous operation for eight (8) hours at plus 140°F. The Modules shall also be capable of continuous operation for eight (8) hours at minus 50°F. During and subsequent to the above temperature extremes (140°F and -50°F), the Modules shall not have been damaged or adversely affected and shall meet the respective Module requirements of 3.5. at least once at the beginning, middle, and ending of each eight (8) hour temperature extreme.

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3.4 Electrical.

3.4.1 Continuity. - The Modules shall demonstrate continuity in accordance with the following respective electrical schematics:

Oscillator, Doppler Gate (A1/A2 Module)	10525365
Generator, Gate (A3 Module)	10525366
Pulse Generator (A4 Module)	10525367
Amplifier, AF/RF (A5/A9 Module)	10525368
Indicator, Reset (A6 Module)	10525369
Oscillator 95.648 KHz, Radio Freq. (A7 Module)	8215498
Gate Delay (A8 Module)	10525371
Oscillator 75 KHz, Radio Freq. (A10 Module)	10525413

3.5 Performance. - The performance requirements specified herein shall be met on every Module at Standard Ambient Conditions prior to government acceptance.

3.5.1 Power Supply - All requirements of 3.5 shall be met within the following power supply ranges except where a paragraph specifically indicates an exception:

(a) +21.56 to+22.44 VDC
(b) +174 to+178 VDC
(c) +10.78 to+11.22 VDC
(d) +5.39 to +5.61 VDC

3.5.2 Oscillator, Doppler Gate (A1/A2 Module) - The Oscillator Doppler Gate shall be capable of producing a 10 KHz to 120 KHz rectangular wave Output on Pin 7 (GATED OUTPUT) swinging between the levels of plus 5.6 $\pm$ 1.3 VDC to plus 11.5 $\pm$ 0.7 VDC with a fall time measured between the 90% and 10% amplitude of less than 0.04 microseconds as shown in FIG I(A). The Output on Pin 7 (GATED OUTPUT) shall be obtained with a plus 6.9 $\pm$ 0.1 VDC applied to Pin 4 (GATE INPUT) and an Input applied to Pin 1 (OSC/DOPPLER INPUT) having a frequency of 10 KHz to 120 KHz with waveshape characteristics as shown in FIG I(B) for the respective temperature values at which the requirement is being met.

3.5.2.1 The Oscillator Doppler Gate shall have a constant Output on Pin 7 (GATED OUTPUT) of either plus 5.6 $\pm$ 1.3 VDC or plus 11.5 $\pm$ 0.7 VDC. The Output on Pin 7 shall be at a constant voltage level and shall not be triggering. The Output on Pin 7 (GATED OUTPUT) shall be obtained with a plus 4.5 $\pm$ 0.1 VDC applied to Pin 4 (GATE INPUT) and an Input applied to Pin 1 (OSC/DOPPLER INPUT) having a frequency of 10 KHz to 120 KHz with waveshape characteristics as shown in FIG I (B) except that the amplitude (AMP) as defined in FIG I(B) shall be reduced to 75% of the value called for in FIG I(B) for the respective temperature value at which the requirement is being met.

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3.5.3.3 The Oscillator Doppler Gate shall have an Output on Pin 7 (GATED OUTPUT) of plus 11.5 ± 0.7 VDC when a plus 11.5 ± 0.7 VDC is applied to Pin 4 (GATE INPUT) and an Input is applied to Pin 1 (OSC/DOPPLER INPUT) as defined in para 3.5.2 with an amplitude (AMP) of 6.9 ± 0.1 volts at all temperature values. Except for the change in amplitude (AMP) at all temperature values, the Input applied to Pin 1 (OSC/DOPPLER INPUT) shall be as shown in FIG I(B).

3.5.3 Generator, Gate (A3 Module) - The Generator Gate shall be capable of changing the Output on Pin 7 (GATE OUTPUT) from plus 18.2 ± 1.9 VDC to plus 6.0 ± 0.6 VDC when an Input is applied to Pin 1 (PULSE INPUT) having pulse characteristics as shown in FIG II.

3.5.3.1 The Generator Gate shall be capable of changing the Output on Pin 7 (GATE OUTPUT) from plus 6.0 ± 0.6 VDC to plus 18.2 ± 1.9 VDC when an Input is applied to Pin 4 (SET INPUT) having pulse characteristics as shown in FIG III.

3.5.3.2 When the plus 5.5 VDC is removed from Pin 2 (RESET), the Output on Pin 7 (GATE OUTPUT) shall change from plus 6.0 ± 0.6 VDC to plus 18.2 ± 1.9 VDC. When the plus 5.5 VDC is reapplied to Pin 2 (RESET), the Output on Pin 7 (GATE OUTPUT) shall remain at plus 18.2 ± 1.9 VDC.

3.5.4 Pulse Generator (A4 Module) - The Pulse Generator shall be capable of changing the Output on Pin 7 (PULSE OUTPUT) from plus 18.2 ± 1.9 VDC to plus 6.0 ± 0.6 VDC when a square wave having a frequency of 1.0 ± 0.05 KHz and waveshape characteristics as shown in FIG IV is applied to Pin 1 (AUDIO INPUT).

3.5.4.1 When the plus 5.5 VDC is removed from Pin 2 (RESET), the Output on Pin 7 (PULSE OUTPUT) shall change from plus 6.0 ± 0.6 VDC to plus 18.2 ± 1.9 VDC. When the plus 5.5 VDC is reapplied to Pin 2 (RESET), the Output on Pin 7 (PULSE OUTPUT) shall remain at plus 18.2 ± 1.9 VDC.

3.5.4 Amplifier AF/RF (A5/A9 Module) - The Amplifier AF/RF shall be capable of producing an Output on Pin 7 (MIC/DOPPLER OUTPUT) over the frequency range of 10 KHz to 120 KHz having a waveshape characteristics as shown in FIG V for the respective temperature value at which the requirement is being met. The Output on Pin 7 (MIC/DOPPLER OUTPUT) shall be obtained with the Input on Pin 1 (MIC/DOPPLER INPUT) being a sine wave over the frequency range of 10 KHz to 120 KHz having a peak-to-peak amplitude as given in FIG V for the respective temperature value at which the requirement is being met.

3.5.5 Indicator, Reset (A6 Module) - The Indicator Reset shall have an Output on Pin 3 (+ LAMP OUTPUT) and on Pin 7 (- LAMP OUTPUT) with the voltage applied to Pin 2 (APPLIED VOLTAGE) and to Pin 1 (PULSE INPUT) as shown in Table I.

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PULSE INPUT Pin 1 (tol ± 0.1VDC)	APPLIED VOLTAGE Pin 2 (tol ± 0.3 VDC)	LAMP OUTPUT - Pin 7	LAMP OUTPUT + Pin 3
+ 6.6 VDC	+ 128.8 VDC	+ 129.1 $^{+0}_{-10}$ VDC	+ 129.1 $^{+0}_{-10}$ VDC
+ 5.4 VDC	+ 128.8 VDC	+ 129.1 $^{+0}_{-10}$ VDC	+ 129.1 $^{+0}_{-10}$ VDC
+ 6.6 VDC	+ 101.0 VDC	+ 101.3 $^{+0}_{-10}$ VDC	+ 101.3 $^{+0}_{-10}$ VDC
+ 5.4 VDC	+ 101.0 VDC	+ 101.3 $^{+0}_{-10}$ VDC	+ 101.3 $^{+0}_{-10}$ VDC
+ 20.2 VDC	+ 116.0 VDC	+ 6 ± 0.6 VDC	+ 46.6 ± 13.2 VDC
+ 16.1 VDC	+ 116.0 VDC	+ 6 ± 0.6 VDC	+ 46.6 ± 13.2 VDC

TABLE I

3.5.6 Oscillator 95.648, Radio Freq. (A7 Module) - The Oscillator (95.648 KHz) shall be capable of producing a 95.648 KHz ± .012% KHz square wave swinging between the levels of plus 6.0 ± 0.6 VDC to plus 11.5 ± 0.7 VDC at Pin 7 (OUTPUT) with a 95.648 KHz Crystal (PART NO. 8215505) across Pins 4 and 9. The crystal shall be mounted in the crystal holder (PART NO. 10525355).

3.5.7. Gate Delay (A8 Module) - The Gate Delay shall be capable of having an Output (DELAYED OUTPUT) on Pin 7 and an Output (SET OUTPUT) on Pin 3 of either plus 11.5 ± 0.7 VDC or plus 5.7 ± 1.3 VDC as given in Table II with Input voltages on Pin (LOGIC #1), Pin 2 (LOGIC #2), and Pin 3 (LOGIC #3) in combinations as given in Table II.

COMBINATION	INPUTS PIN NO.	VOLTAGES (tol ± 0.1 VDC)	OUTPUT PIN 3	OUTPUT PIN 7
	1	+ 5.3 VDC	+ 11.5 ± 0.7 VDC	+ 5.7 ± 1.3 VDC
A	2	+ 5.3 VDC		
	4	+ 5.3 VDC		
	1	+ 12.3 VDC	+ 5.7 ± 1.3 VDC	+ 11.5 ± 0.7 VDC
B	2	+ 6.7 VDC		
	4	+ 5.3 VDC		
	1	+ 5.3 VDC	+ 5.7 ± 1.3 VDC	+ 11.5 ± 0.7 VDC
C	2	+ 11.5 VDC		
	4	+ 6.7 VDC		
	1	+ 6.7 VDC	+ 5.7 ± 1.3 VDC	+ 11.5 ± 0.7 VDC
D	2	+ 5.3 VDC		
	4	+ 10.7 VDC		
	1	+ 6.7 VDC	+ 11.5 ± 0.7 VDC	+ 5.7 ± 1.3 VDC
E	2	+ 6.7 VDC		
	4	+ 6.7 VDC		

TABLE II

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3.5.8 Oscillator 75 KHz, Radio Freq. (A10 Module) - The 75 KHz Radio Frequancy Oscillator shall be capable of producing a 75 KHz ± .012% KHz semi-sawtooth wave with a peak-to-peak amplitude of 0.6 ± 0.2 volts at the Pin 7 (OUTPUT) with a 75 KHz Crystal (PART NO. 10525357) across Pins 1 and 3. The crystal shall be mounted in the crystal holder (PART NO. 10525355)

3.6 Reliability Run-in. - In order to maximize component reliability, prior to acceptance by the government, each and every Module shall be subject to the following run-in conditions:

3.6.1 Temperature Run-in. - The Module, without power applied, shall be temperature stabilized at plus 125°F and minus 25°F. At each temperature extreme, after temperature stabilization, power shall be applied and each Module shall be tested to the respective Module requirements of 3.5 for a period of one (1) hour with a minimum of three (3) complete tests to the respective Module re-requirements of 3.5 occurring at the beginning, middle and end of the one (1) hour period. Subsequent to the above tests, the Module shall not have been damaged or adversely affected, and shall meet the respective Module requirements of 3.5.

3.6.2. Vibration Run-in. - The Module shall be capable of continuous operation while being vibrated in a simple harmonic motion having an amplitude of 0.03 (0.06 inch maximum total excursion) with the frequency being varied uniformly between the limits of ten (10) and fifty-five (55) hertz and return to ten (10) hertz and shall be uniformly traversed in one (1) minute. This motion shall be applied for a period of ten (10) minutes in each of three (3) mutally perpendicular directions (x, y, z axis) for a total of 0.5 hours. During the above vibration, the Module shall meet the respective Module performance 3.5 once in each plane. Subsequent to the above vibration, the Module shall not have been damaged or adversely affected, and shall meet the respective Module requirements of 3.5.

3.6.3 Shock Run-in. - The Module shall be capable of three (3) shocks of 50 gravity units (g's) of eleven (11) millisecond duration in each of the following directions:

(a) Bottom down
(b) Front down
(c) Right side down
(d) Back down
(e) Left side down
(f) Top down

Subsequent to the above shock, the Module shall not have been damaged or adversely affected, and shall meet the respective Module requirements of 3.5.

3.6.4 Failures During Reliability Run-in. - Replacement of defective components and correction of workmanship defects may be made during Run-In under the following conditions:

(a) Any part, module, component or subassembly that is used as a replacement shall have previously passed environmental tests consistent with the Reliability Run-In environmental requirements of the respective Module.

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(b) Any replacement part shall have been exposed for at least one (1) hour at the extreme temperature being run immediately previous to being used as a replacement for any failure occurring during a temperature extreme.

(c) After a correction has been accomplished, at least one (1) hour of operation with a minimum of three (3) tests to the respective Module requirements of 3.5 shall be sucessfully accomplished at the temperature at which the failure occurred.

(d) A failure during paragraphs 3.6.2. or 3.6.3 shall necessitate a complete re-run of the conditions of paragraphs 3.6.2 and 3.6.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the Contract or Purchase Order, the Supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the Supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General Provisions - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First Article - Testing and Approval - The requirements for First Article Approval and the responsibility (Government or Contractor) for First Article Testing shall be as specified in the Contract. The sample for First Article Approval Tests shall consist of three (3) Modules manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government Testing - When the Government is responsible for conducting First Article Testing (as specifically delineated in the Contract), the Contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the Contract and submit a record of this inspection with the sample, including Certificates of Conformance for materials.

4.2.2 Contractor Testing - When the Contractor is responsible for conducting First Article Testing (as per para 4.2), the sample shall be inspected by the Contractor for all the requirements of the Contract. The sample and a record of this inspection, including Certificates of Conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the Contractor's inspection.

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4.3 Inspection Provisions.

4.3.1 Examination and Test.

4.3.2 Acceptance and Rejection - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special Sampling - Subsequent to First Article Approval (see 4.2), three (3) units shall be selected at random by a Government representative from the first fifty (50) units, or portion thereof, produced and two (2) additional units from each subsequent fifty (50), or portion thereof, produced. The Special Sample units shall have successfully met all examinations and acceptance tests specified herein; and no lots from which the Special Sample units have been selected shall be accepted by the Government until the Special Sample units have successfully met the following requirements.

	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
401.	Fabrication	3.1	Applic dwgs & Specs - Visual
402.	General Specifications	3.2	MIL-F-13926 & MIL-C-46357
403.	Shock	3.3.1	4.6.1.1
404.	Vibration	3.3.2	4.6.1.2
405.	Humidity	3.3.3	4.6.1.3
406.	Sand and Dust	3.3.4	4.6.1.4
407.	Altitude-operating	3.3.5.1	4.6.1.5
408.	Altitude-non-operating	3.3.5.2	4.6.1.5.1
409.	Storage temperatures	3.3.7	4.6.1.7.1
410.	Operating temperatures	3.3.8	4.6

4.4.1 Failure of Special Sample - Should any one unit of a Special Sampling (see 4.4) fail to meet the specified test requirements, necessary corrections shall be made by the Contractor to the lot of units from which the Special Samples were selected and reselected Special Samples per paragraph 4.4 shall be submitted and tested for approval.

4.5 Inspection equipment - Except as otherwise provided for by the Contract, the Contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government Furnished Inspection Equipment.- Where the Contract provides for Government-Furnished Test Equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor Furnished Inspection Equipment.

4.5.2.1 Contractor Design.- The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification, and with the "Component Inspection" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of product which does not exceed 90% of the prescribed tolerance limit and permit positive rejective when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test Methods Procedures.

4.6.1 Environmental.- Test equipment and test facilities utilized in the following environmental tests shall be in accordance with "Test Facilities" requirements of MIL-F-13926 and conditions specified in 3.3.

4.6.1.1 Shock.- The shock test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV at the conditions specified in 3.3.1. Subsequent to the eighteen (18) shocks, the Module shall be visually examined for damage, then tested as specified in 4.6.2.

4.6.1.2 Vibration.- The vibration test shall be performed as specified in MIL-STD-810, Method 514 at the conditions specified in 3.3.2. During the above vibration, the module shall meet the performance test as specified in 3.3.2. Subsequent to the 1.5 hour vibration, the Module shall be visually examined for damage, then tested as specified in 4.6.2.

4.6.1.3 Humidity.- The humidity test shall be performed as specified in MIL-STD-810, Method 507, Procedure I for four (4) hours at the conditions specified in 3.3.3. Subsequent to this test, the Module shall be examined for damage then tested as specified in 4.6.2.

4.6.1.4 Sand and Dust.- The sand and dust test shall be performed as specified in MIL-STD-810, Method 510, with the following conditions: The sand and dust particle sizes, density, wind velocity, temperature, time duration and relative humidity shall be as specified in 3.3.4. Subsequent to this test, the module shall be visually examined for damage then tested as specified in 4.6.2.

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4.6.1.5 Altitude-Operating - The altitude test shall be performed as specified in MIL-F-13926, Procedure I at an altitude of 10,000 feet and as specified in 3.3.5.1. The Module shall be exposed to the above condition for a time period required to allow all entrapped air in the chamber to escape as evidenced by chamber pressure stabilization. The barometric pressure shall then be maintained to simulate 10,000 feet (20.58 inches of mercury) altitude condition. While the Module is maintained at this pressure for four (4) hours, the Module shall be subjected to the test specified in 4.6.2.

4.6.1.5.1 Altitude-Non-Operating- The Module shall be subjected to barometric pressure specified in MIL-F-13926, Procedure II for four (4) hours as specified in 3.3.5.2. Subsequent to this test, the Module shall be visually examined for damage, then tested as specified in 4.6.2.

4.6.1.6 Fungus - One (1) Module shall be subjected to the fungus test specified in MIL-F-13926. At the end of the 28 day incubation period, the Module shall be tested as specified in 4.6.2, for compliance with 3.3.6. Subsequent acceptance of Module shall be by certification.

4.6.1.7 Temperature Cycling.

4.6.1.7.1 Storage Temperatures - This test shall be performed as specified in MIL-F-13926, Procedure II, and thermally stabilized at +160°F for eight (8) hours and at -80°F for eight (8) hours. The Module shall be returned to room ambient temperature +75 ± 15°F, visually examined for damage then tested as specified in 4.6.2.

4.6.1.7.2 Operating Temperatures - The Module shall be placed in a test chamber and the temperature gradually raised at a rate not to exceed 2°F per minute unitl thermally stabilized at +140°F for eight (8) hours. When thermally stabilized at this temperature, the Module shall be tested as specified in 4.6.2 during the first hour, at the four (4) hour period, and within the last hour of the eight (8) hour period. The temperature shall then be reduced to -50°F. When thermally stabilized at this temperature, the Module shall be tested as specified in 4.6.2. at the three (3) time periods specified above. The Module shall be returned to room ambient temperature +75 ± 15°F, examined for damage then tested as specified in 4.6.2.

4.6.2 Performance - Unless otherwise specified, the performance tests specified herein shall be satisfactorily accomplished on every Module at Standard Ambient Conditions. Standard Ambient Conditions for the purposes of these tests shall be defined as:

(a) Temperature between 60°F and 90°F.

(b) Barometric pressure between 28 and 32 inches of mercury.

(c) Relative humidity (RH) up to 90%.

4.6.2.1 Initial Conditions - The following voltage and test equipment connections shall be applied as diagrammed in the respective Module Test Set-Up Figures.

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4.6.2.2 Power Supply - All requirements of para 4.6.2. shall be accomplished with the power supply inputs over the ranges given in para 3.5.1.

4.6.2.3 Oscillator, Doppler Gate (A1/A2 Module) - Connection of voltages and test equipment shall be as shown in FIG VI. Place Switch SW1 to position A. Adjust the voltage on Pin 4 (GATE INPUT) to plus 6.9 ± 0.1 VDC. Adjust and monitor the sine wave generator for an Input on Pin 1 (OSC/DOPPLER INPUT) having waveshape characteristics as shown in FIG. I (B). (Amplitude (AMP) shall be as defined for the temperature value at which the test is being conducted.) The frequency of the Input on Pin 1 (OSC/DOPPLER INPUT) shall be 10 KHz ± 0.3 KHz. The Output on Pin 7 (GATED OUTPUT) shall be as shown in FIG. I (A) with a frequency the same as the Input frequency on Pin 1 (OSC/DOPPLER INPUT).

4.6.2.3.1 Repeat the tests of para 4.6.2.3 with the Input frequency on Pin 1 (OSC/DOPPLER INPUT) changed to 120 KHz ± 0.5 KHz. Except for the change in the Input frequency on Pin 1 (OSC/DOPPLER INPUT), all test requirements are as given in para 4.6.2.3.

4.6.2.3.2. Repeat the tests of para 4.6.2.3 and para 4.6.2.3.1. The amplitude (AMP) of the Input on Pin 1 (OSC/DOPPLER INPUT) shall be reduced to 75% of the value defined in FIG I (B) for the temperature value at which the test is being conducted and the Output on Pin 7 (GATED OUTPUT) shall be at a constant plus 5.6 ± 1.3 VDC level or at a constant plus 11.5 ± 0.7 VDC level and shall not be triggering. A plus 4.5 ± 0.1 VDC shall be applied to Pin 4 (GATE INPUT). Except for the change in the amplitude (AMP) of the Input on Pin 1 (OSC/DOPPLER INPUT), the constant voltage Output on Pin 7 (GATED OUTPUT), and the change in the voltage level applied to Pin 4 (GATED INPUT); the tests shall be conducted as defined in para 4.6.2.3 and para 4.6.2.3.1.

4.6.2.3.3 Repeat the tests of para 4.6.2.3 and para 4.6.2.3.1. Place Switch SW1 to position B. The amplitude (AMP) of the Input on Pin 1 (OSC/DOPPLER INPUT) shall be 6.9 ± 0.1 volts at all temperature values defined in FIG I(B) at which the tests are being conducted. The Output on Pin 7 (GATED OUTPUT) shall be plus 11.5 ± 0.7 VDC when a plus 11.5 ± 0.7 VDC is applied to Pin 4 (GATE INPUT) by placing Switch SW1 to position B and adjusting the voltage on Pin 4 (GATED INPUT) to plus 11.5 ± 0.7 VDC.

4.6.2.4 Generator Gate (A3 Module) - Connection of voltages and test equipment shall be as shown in FIG VII. Close Switch SW2 such that plus 5.5 VDC is applied to Pin 2 (RESET). The Output on Pin 7 (GATE OUTPUT) shall be plus 18.2 ± 1.9 VDC. Apply a ONE-SHOT pulse to Pin 1 (PULSE INPUT) having pulse characteristics as shown in FIG II. The Output on Pin 7 (GATE OUTPUT) shall change from plus 18.2 ± 1.9 VDC to plus 6.0 ± 0.6 VDC and shall remain at plus 6.0 ± 0.6 VDC.

4.6.2.4.1 The Output on Pin 7 (GATE OUTPUT) shall be at plus 6.0 ± 0.6 VDC. Apply an Input to Pin 4 (SET INPUT) having pulse characteristics as shown in FIG III. The Output on Pin 7 (GATE OUTPUT) shall change from plus 6.0 ± 0.6 VDC to plus 18.2 ± 1.9 VDC.

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4.6.2.4.2 Repeat the test of para 4.6.2.4. in order to place the Generator Gate in the condition that the Output on Pin 7 (GATE OUTPUT) is at a plus 6.0 ± 0.6 VDC. Open Switch SW2 such that the plus 5.5 VDC is removed from Pin 2 (RESET). The Output on Pin 7 (GATE OUTPUT) shall change from plus 6.0 ± 0.6 VDC to plus 18.2 ± 1.9 VDC. Close Switch SW2 such that the plus 5.5 VDC is reapplied to Pin 2 (RESET). The Output on Pin 7 (GATE OUTPUT) shall be plus 18.2 ± 1.9 VDC and shall remain at plus 18.2 ± 1.9 VDC.

4.6.2.5. Pulse Generator (A4 Module) - Connection of voltages and test equipment shall be as shown in FIG VIII. Close Switch SW3 such that plus 5.5 VDC is applied to Pin 2 (RESET). The Output on Pin 7 (PULSE OUTPUT) shall be plus 18.2 ± 1.9 VDC. Adjust and monitor the Square Wave Generator to the requirements called for in para 3.5.4. and FIG IV. Close Switch SW4. The Output on Pin 7 (PULSE OUTPUT) shall change from plus 18.2 ± 1.9 VDC to plus 6.0 ± 0.6 VDC. Open Switch SW4. The Output on Pin 7 (PULSE OUTPUT) shall remain at plus 6.0 ± 0.6 VDC.

4.6.2.5.1 The Output on Pin 7 (PULSE OUTPUT) shall be at plus 6.0 ± 0.6 VDC. Open Switch SW3 such that the plus 5.5 VDC is removed from Pin 2 (RESET). The Output on Pin 7 (PULSE OUTPUT) shall change from plus 6.0 ± 0.6 VDC to plus 18.2 ± 1.9 VDC. Close Switch SW3 such that the plus 5.5 VDC is reapplied to Pin 2 (RESET). The Output on Pin 7 (PULSE OUTPUT) shall be plus 18.2 ± 1.9 VDC and shall remain at plus 18.2 ± 1.9 VDC.

4.6.2.6. Amplifier, AF/RF (A5/A9 Module) - Connection of voltages and test equipment shall be as shown in FIG IX. Adjust and monitor the Sine Wave Generator for a 10 KHz ± 0.3 KHz sine wave having a peak-to-peak amplitude of 100 millivolts when applied as an Input to Pin 1 (MIC/DOPPLER INPUT). The Output on Pin 7 (MIC/DOPPLER OUTPUT) shall be a Clipped Sine Wave with a frequency of 10 KHz ± 0.5 KHz having an amplitude and waveshape characteristics at the temperature value at which the test is being conducted as shown in FIG V.

4.6.2.6.1 Repeat the tests of para 4.6.2.6. with the Input frequency on Pin 1 (MIC/DOPPLER INPUT) and the Output frequency on Pin 7 (MIC/DOPPLER OUTPUT) changed to 120 KHz ± 0.5 KHz. Except for the change in the Input frequency On Pin 1 (MIC/DOPPLER INPUT) and the Output frequency on Pin 7 (MIC/DOPPLER OUTPUT), all test requirements are given in para 4.6.2.6.

4.6.2.6.2 Repeat the tests of para 4.6.2.6. and para 4.6.2.6.1 with the sine wave Input to Pin 1 (MIC/DOPPLER INPUT) reduced to 50 millivolts peak-to-peak. Except for the change in the sine wave amplitude applied to Pin 1 (MIC/DOPPLER INPUT), all test requirements are as given in para 4.6.2.6 and para 4.6.2.6.1.

4.6.2.7 Indicator, Reset (A6 Module) - Connection of voltages and test equipment shall be as shown in FIG X. Adjust and monitor the D.C. power supply on Pin 2 (APPLIED VOLTAGE) and on Pin 1 (PULSE INPUT) to the requirements of TABLE I. The corresponding voltages measured on Pin 3 (+ LAMP OUTPUT) and on Pin 7 (- LAMP OUTPUT) shall be as given in Table I.

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4.6.2.8 Oscillator 95.648, Radio Freq. (A7 Module) - Connection of voltages and test equipment shall be as shown in FIG XI. The Output on Pin 7 (OUTPUT) shall be a 95.648 KHz ± .012% KHz square wave swinging between the levels of plus 11.5 ± 0.7 VDC and plus 6.0 ± 0.6 VDC.

4.6.2.9 Gate Delay (A8 Module) - Connection of voltages and test equipment shall be as shown in FIG XII. Adjust and monitor the Inputs (LOGIC #1, LOGIC #2 LOGIC #3) in Combinations (A, B, C, D, E) as given in Table II such that the Outputs (SET OUTPUT, DELAYED OUTPUT) correspond to the respective Input Combination as given in Table II.

4.6.2.10 Oscillator 75 KHz, Radio Freq. (A10 Module) - Connection of voltages and test equipment shall be as shown in FIG XIII. The Output on Pin 7 (OUTPUT) shall be a 75 KHz ± .012% KHz semi-sawtooth wave with a peak-to-peak amplitude of 0.6 ± 0.2 volts as shown in FIG XIII.

4.6.3 Electrical

4.6.2.1 Continuity - Continuity shall be tested by standard measuring equipment for compliance with 3.4.1.

4.6.4. Reliability Run-in - The following run-in tests shall be performed on a 100% basis on each and every Module.

4.6.4.1 Temperature Run-in - The Module shall be placed in a chamber and the temperature gradually raised at a rate not to exceed 2°F per minute until thermally stabilized at plus 125°F. After temperature stabilization, power shall be applied and maintained for one hour. Five minutes after power is applied, a complete test to the respective Module requirements of para 4.6.2 shall commence. A second complete test to the respective Module requirements of para 4.6.2 shall be accomplished during the middle of the hour and a third test to the respective Module requirements of para 4.6.2 shall be accomplished during the last part of the hour test. Repeat the above tests at a temperature of minus 25°F. Subsequent to the above temperature extremes (plus 125°F and 25°F) the Module shall not have been damaged or adversely affected and shall meet the respective Module requirements of para 4.6.2.

4.6.4.2 Vibration Run-in - The vibration test shall be performed as specified in MIL-STD-810, Method 514 at the conditions specified in 3.6.2. During the above vibration test the Module shall meet the performance tests as specified in para 3.6.2. Subsequent to the vibration test, the Module shall be visually examined for damage and shall not be damaged or adversely affected and shall meet the respective Module requirements of para 4.6.2.

4.6.4.3 Shock Run-in - The shock test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV at the conditions specified in 3.6.3. Subsequent to the eighteen (18) shocks, the Module shall be visually examined for damage and shall not be damaged or adversely affected and shall meet the respective Module requirements of para 4.6.2

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4.6.4.4 Failures - Failures during Reliability Run-in shall not be cause for rejection, but shall be corrected as outlined in para 3.6.4.

4.6.5. Packaging, Packing and Marking - Quality Assurance Provisions for packaging, packing and marking shall be as specified in MIL-P-14232 and Section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking shall be in accordance with MIL-P-14232 and the appropriate Packaging Data Sheet per para 2.1. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use - The Modules are used as subassemblies in the M36 Radar Chronograph.

6.2. Ordering Data - Procurement documents should specify the following:

(a) Title, Number, and date of this specification.

(b) Selection of applicable levels of preservation, packaging and packing.

(c) Applicable stock number

(d) Applicable packaging data sheet (see 5.1).

6.3 Definitions.- The term Module referenced in this specification refers to each respective module under contract.

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Preparing activity:

Army - MU

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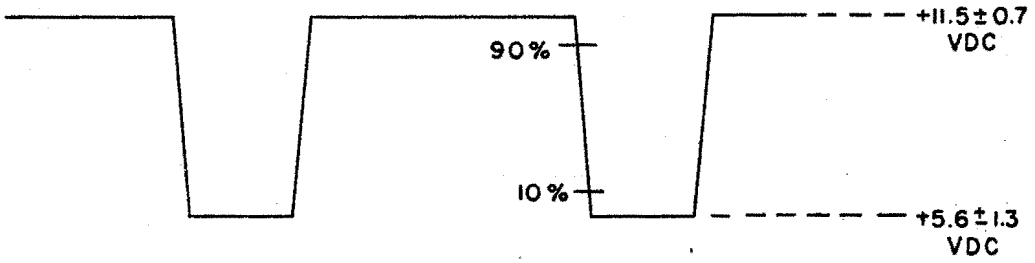
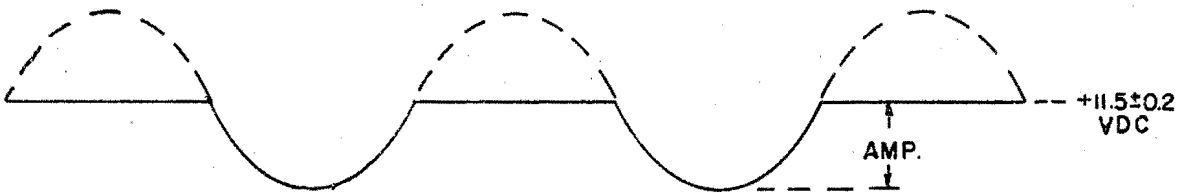


FIG. I (A)



AMP = 3.7 ± 0.2 volts at 70°F ± 10°F

AMP = 3.0 ± 0.2 volts at -25°F

AMP = 2.5 ± 0.2 volts at -50°F

AMP = 4.0 ± 0.2 volts at +125°F

AMP = 4.7 ± 0.2 volts at +140°F

CLIPPED SINE WAVE

FIG. I (B)

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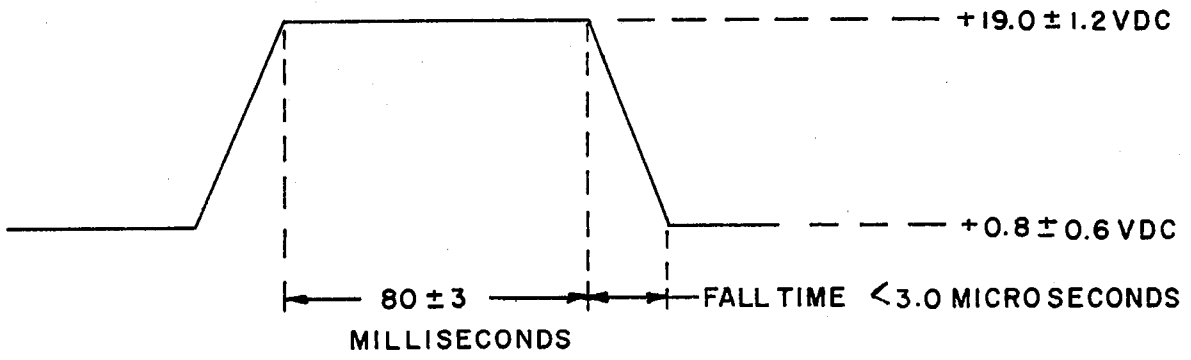


FIG. II

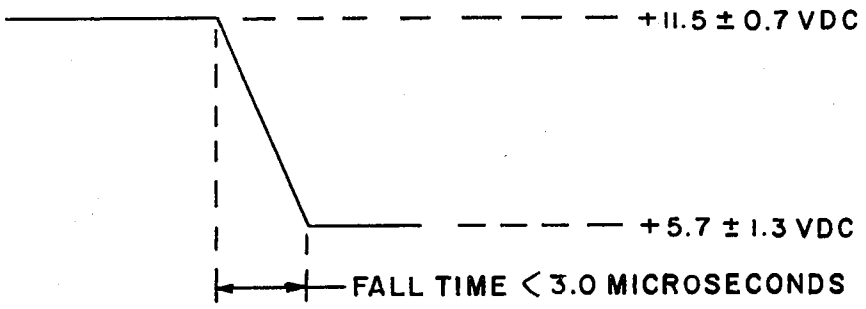
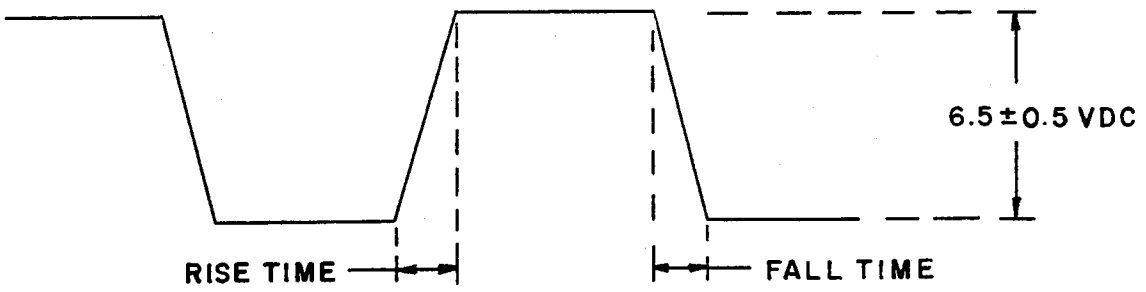


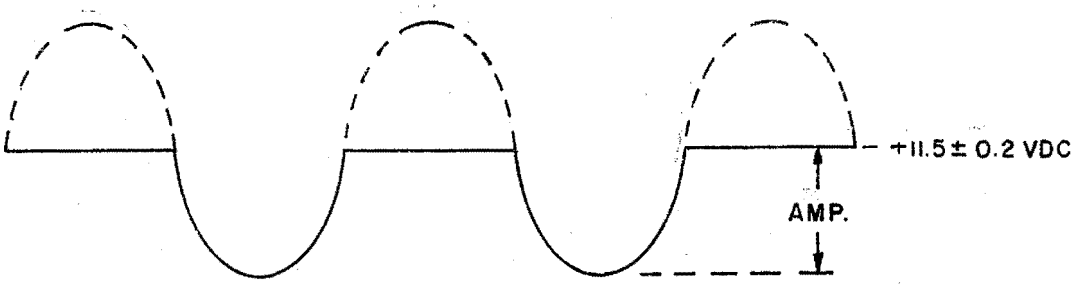
FIG. III



RISE AND FALL TIMES < 0.04 MICROSECONDS

FIG. IV

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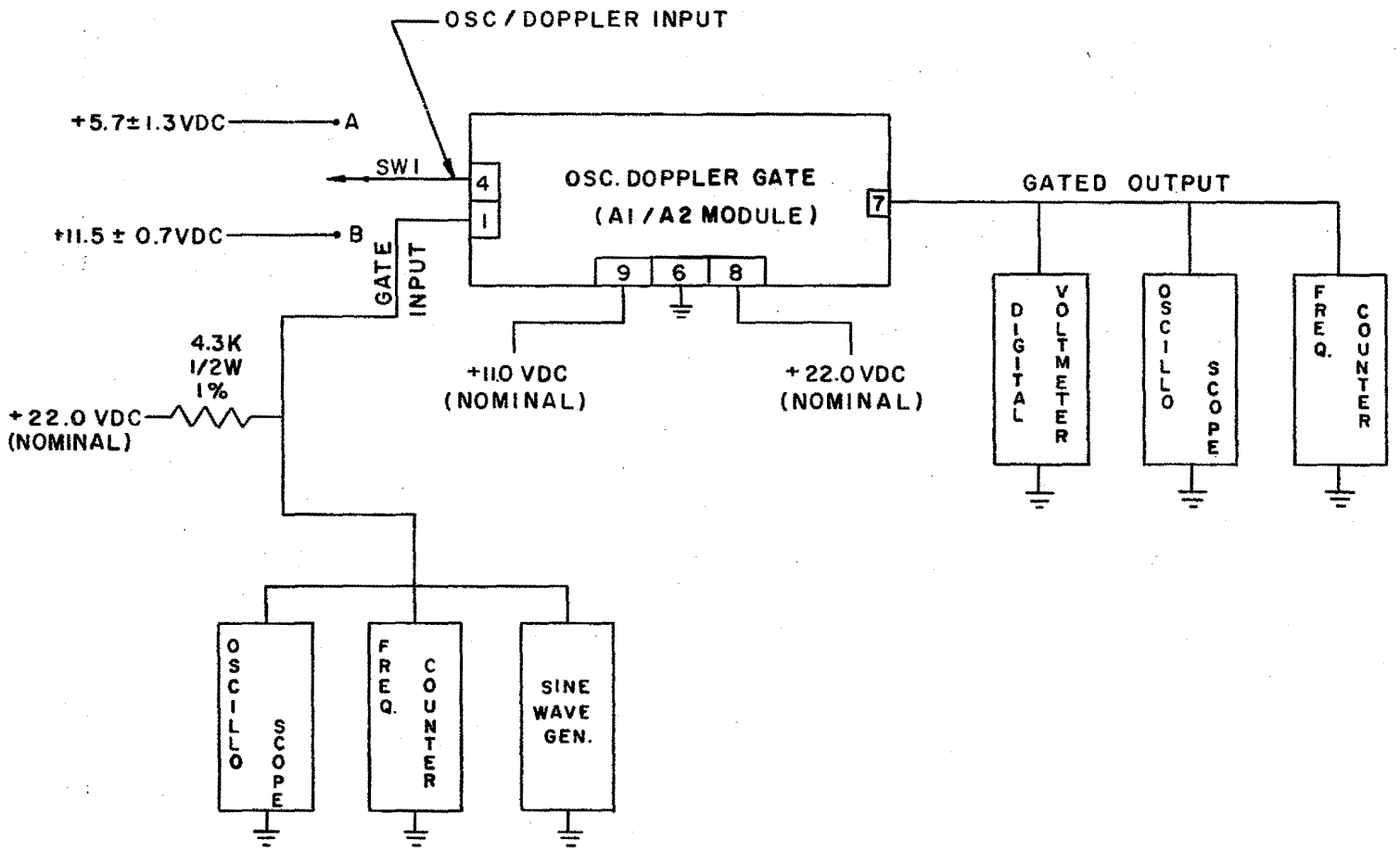


INPUT SINE WAVE PEAK-TO-PEAK	OUTPUT AMPLITUDE (AMP)	TEMPERATURE VALUE
100 Millivolts	>4.5 ± 0.2 Volts	70°F ± 10°F
50 Millivolts	<3.0 ± 0.2 Volts	70°F ± 10°F
100 Millivolts	>3.7 ± 0.2 Volts	-25°F
50 Millivolts	<2.5 ± 0.2 Volts	-25°F
100 Millivolts	>3.0 ± 0.2 Volts	-50°F
50 Millivolts	<1.8 ± 0.2 Volts	-50°F
100 Millivolts	>4.8 ± 0.2 Volts	+125°F
50 Millivolts	<3.2 ± 0.2 Volts	+125°F
100 Millivolts	>5.2 ± 0.2 Volts	+140°F
50 Millivolts	<4.0 ± 0.2 Volts	+140°F

CLIPPED SINE WAVE

FIG. V

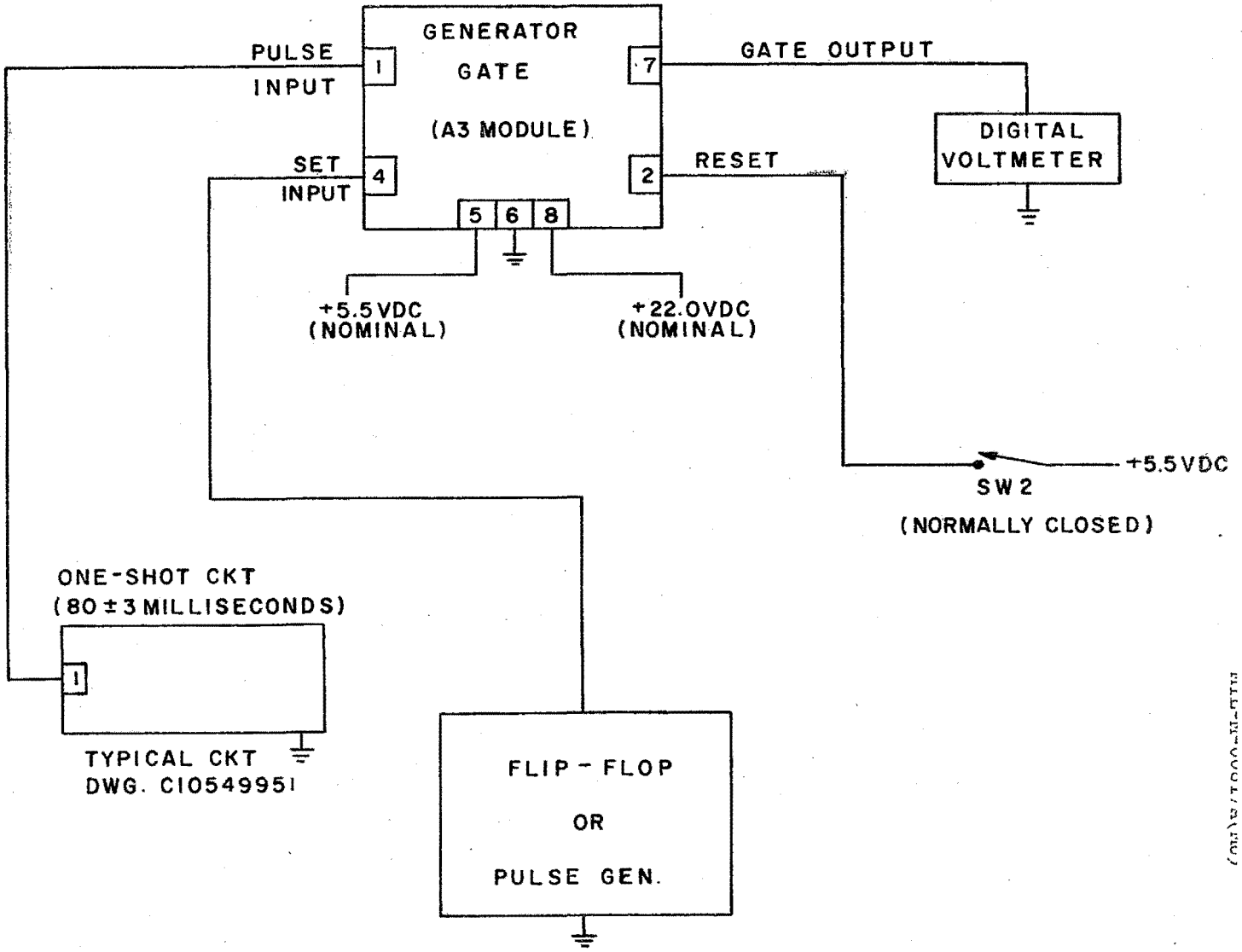
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A1/A2 MODULE TEST SET-UP

FIG. VI

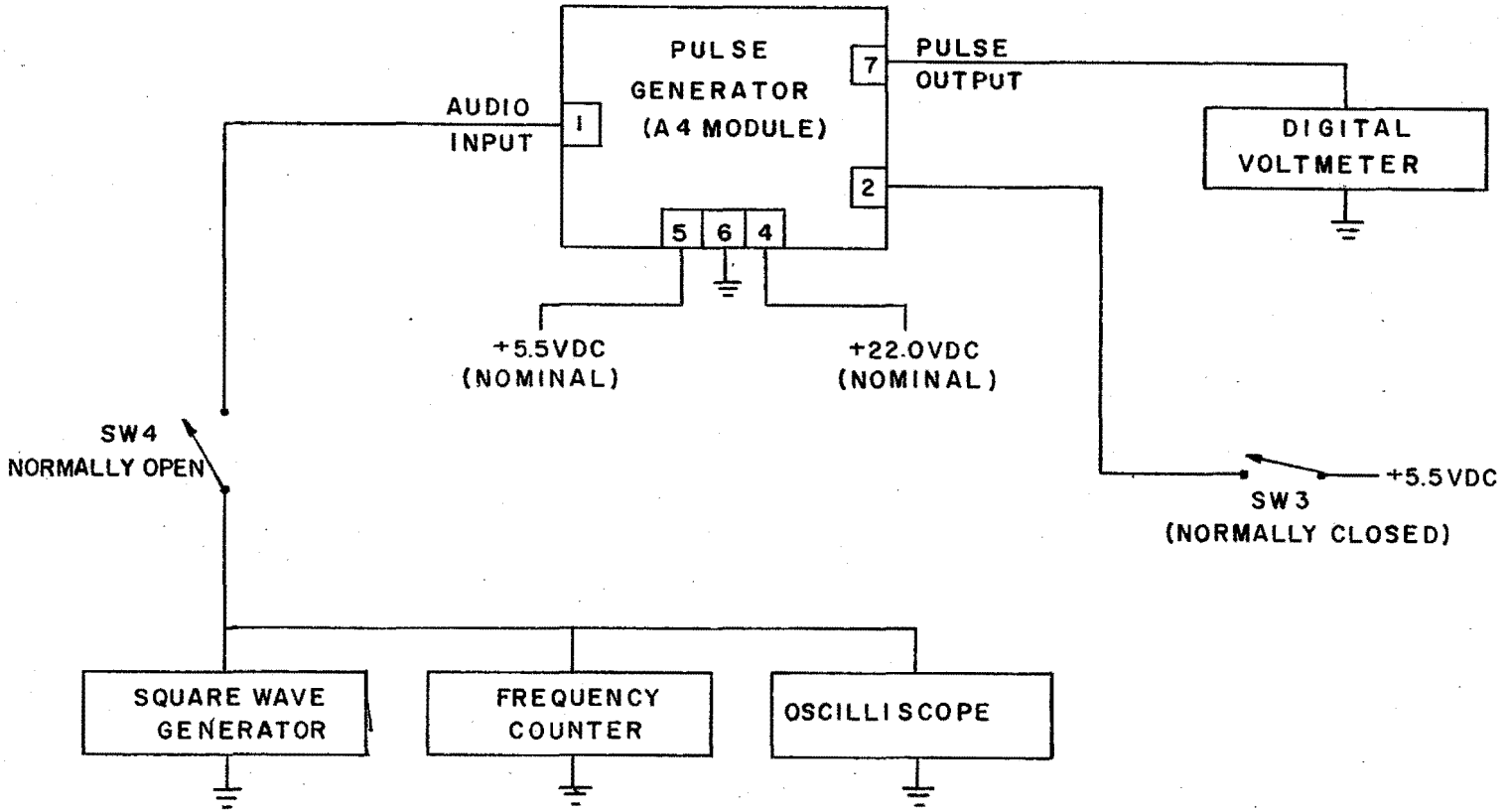
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A3 MODULE TEST SET-UP

FIG. VII

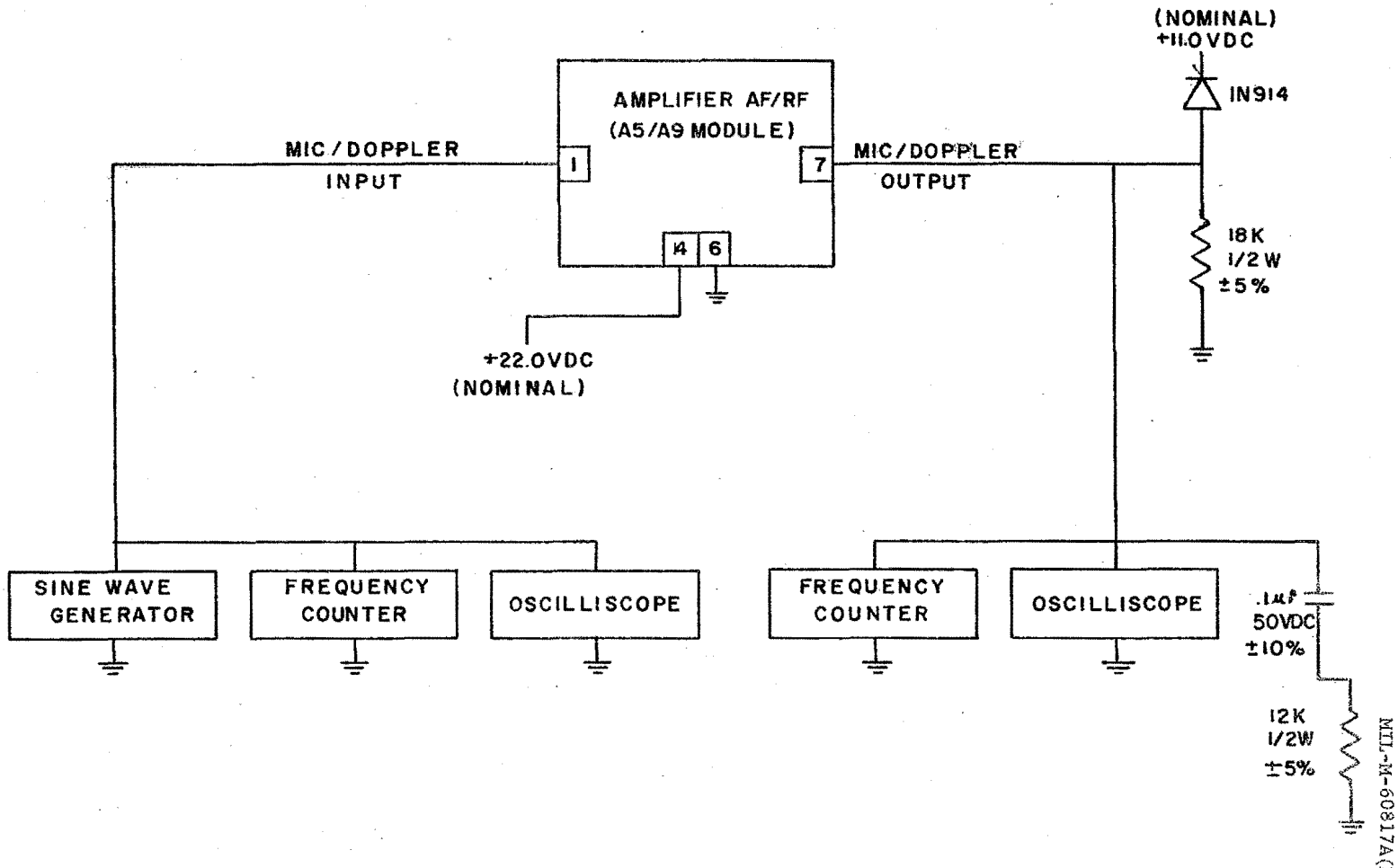
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A4 MODULE TEST SET UP

FIG. VIII

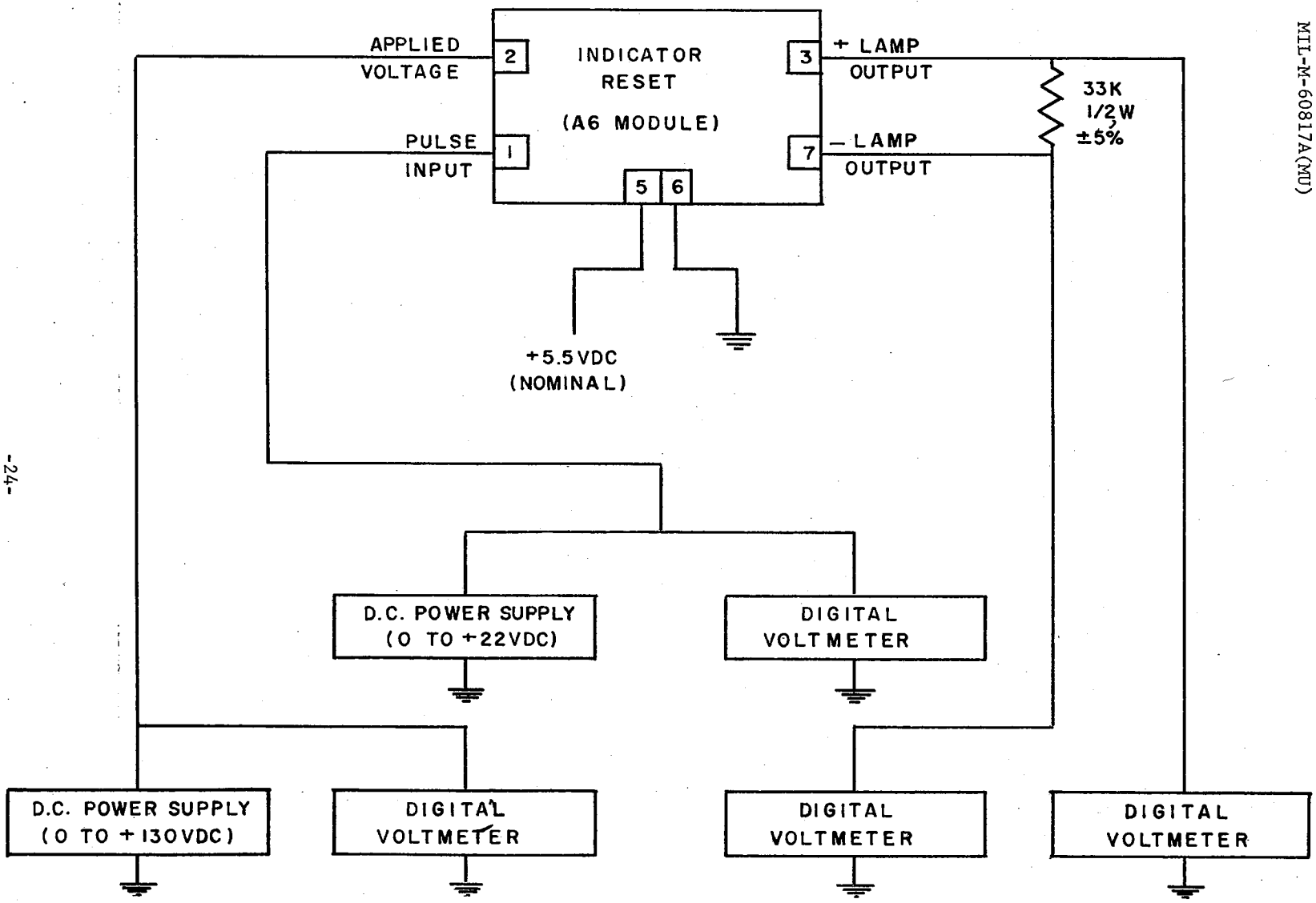
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A5/A9 MODULE TEST SET-UP

FIG. IX

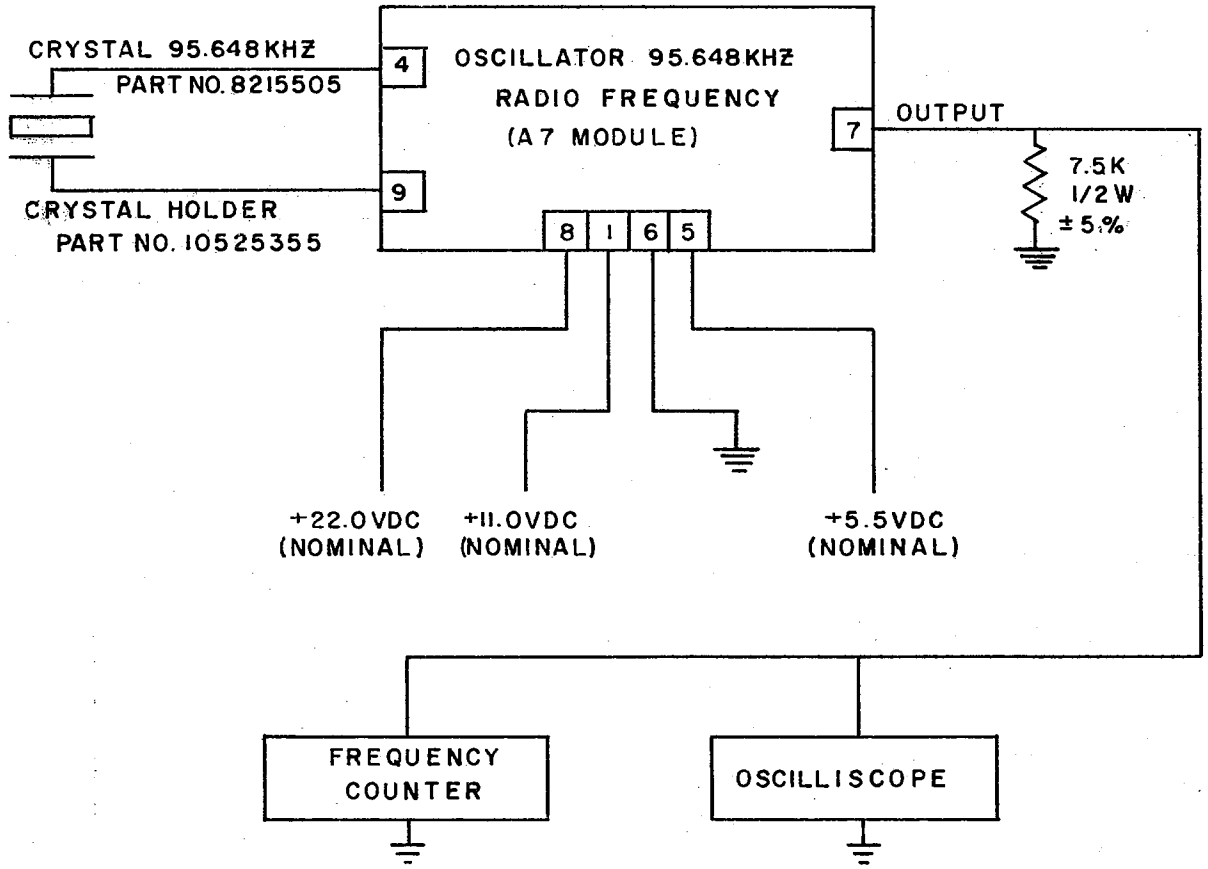
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A6 MODULE TEST SET-UP

FIG. X

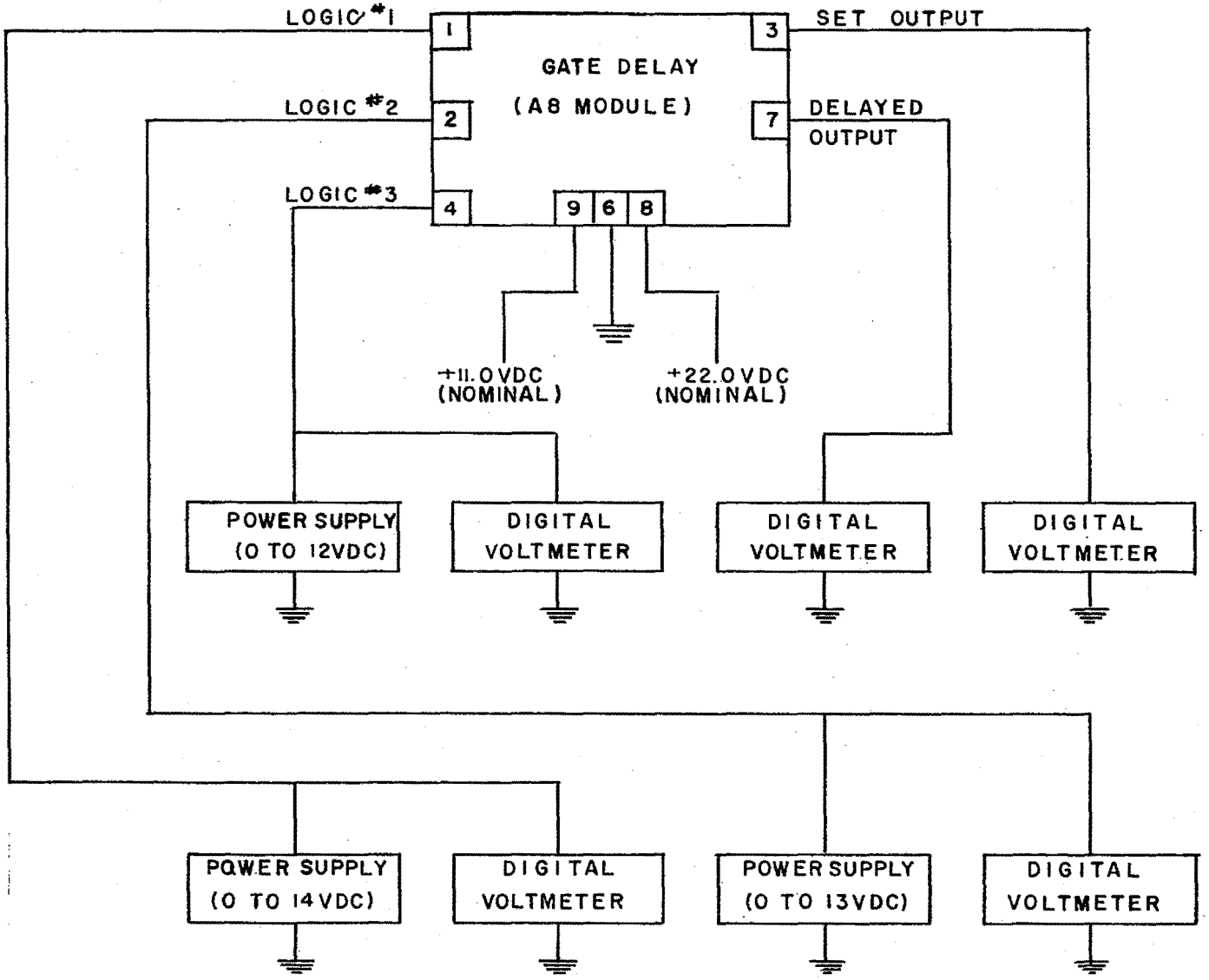
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A7 MODULE TEST SET-UP

FIG. XI

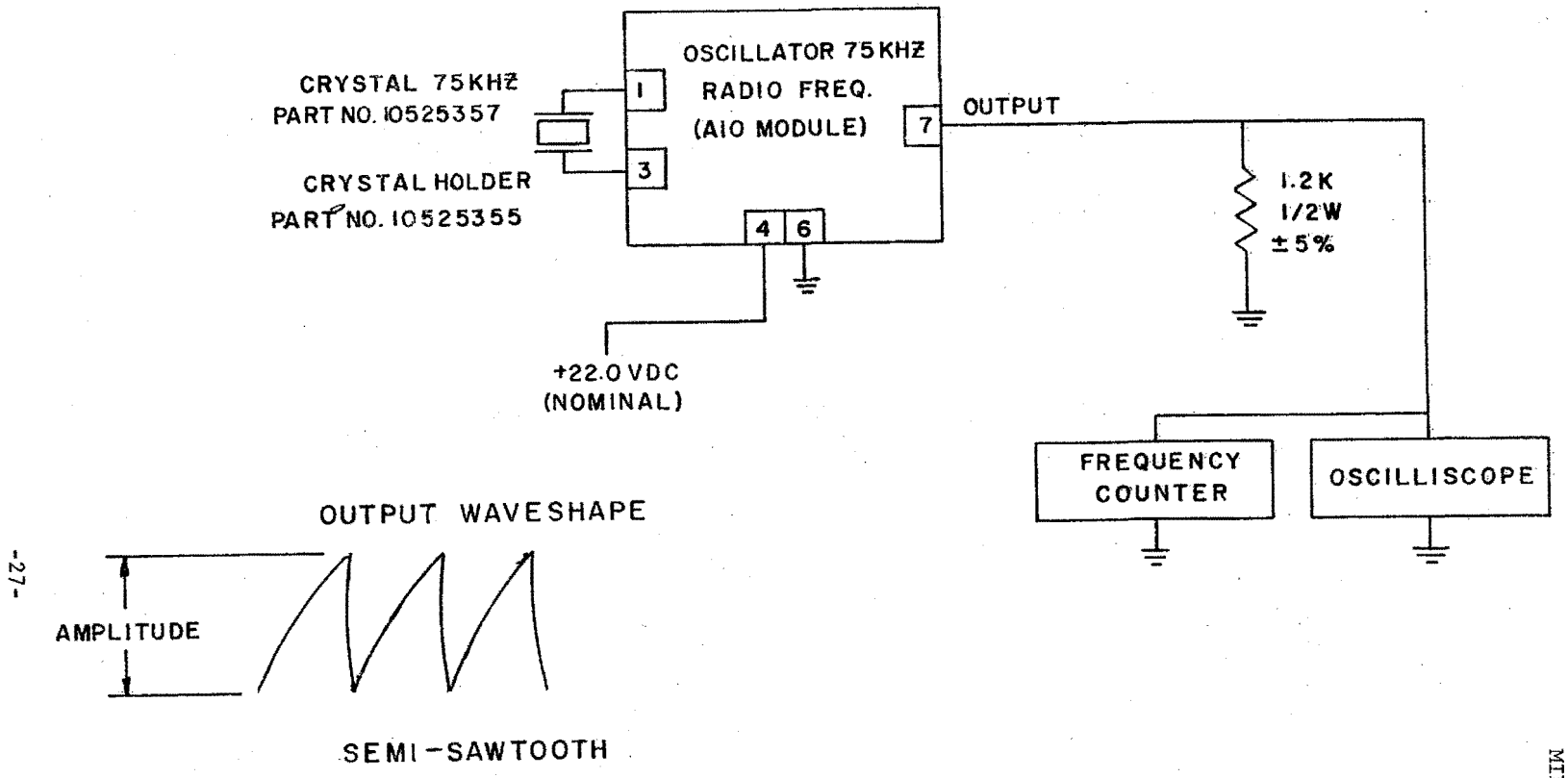
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A8 MODULE TEST SET-UP

FIG. XII

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AIO MODULE TEST SET-UP

FIG. XIII

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